CS 3563: Introduction to DBMS II

Homework 5: Transaction Management

Due Date and Time: 26 March 2026, 11:59 PM IST

Instruction *Total Marks: 25*

1. The homework must be completed individually.
2. Referring to external resources beyond the class notes and reference textbooks/papers is not permitted.
3. The submission must be a single PDF and submitted via Moodle.
4. Late submissions will be handled in accordance with the course policy.

Question 1 **(8 Marks)**

The local branch manager of State Bank of India informs you that all transactions at their branch are composed of the following four basic operations:

Operation	Meaning
$R(a, v)$	Returns in v the current balance in account number a
$S(a, v)$	Sets the balance in account number a to be value v
$D(a, v)$	Deposits amount v to balance in account number a
$W(a, v)$	Withdraws amount v from balance in account number a

Given that a separate lock is available for each operation, draw a lock compatibility matrix. Justify your matrix entries.

Question 2 **(6 Marks)**

Transactions $T1$, $T2$ and $T3$ arrive in that order to a DBMS and are executed concurrently according to the (partial) schedule shown below. ($Ri(x)$ means that transaction i reads data item x and $Wi(x)$ means that transaction i writes data item x.)

$$R2(d)R1(a)W1(a)R3(b)R3(c)R2(a)W2(a)R2(c)W3(c)R1(d)R3(d)W1(d)\ldots$$

Based on this schedule, determine whether it is conflict-serializable. Can 2PL produce this schedule?

Question 3 **(6 Marks)**

You have implemented a log-based recovery manager for a database system. However, your assignment partner recommends that you remove it since they have been given a guarantee by the vendor that the computer hardware and software will never fail – that is, the database system is permanently operational. Assuming that the guarantee is valid, how would you respond to their request? Explain. Would your answer change based on whether the DBMS implemented immediate updates to disk or deferred updates (update to disk only at commit)?

Question 4 **(5 Marks)**

Your classmate implements the following log-based recovery manager: Log records of data updates are always “flushed” (i.e. synchronously written to the log disk) before the corresponding updates are written to disk. At transaction commit time, any data updates that are still memory-resident are written to disk. The transaction commit record is then written to the log disk. Does your classmate’s system provide recoverability? If yes, what will the undo and redo operations do during recovery? If no, explain why?